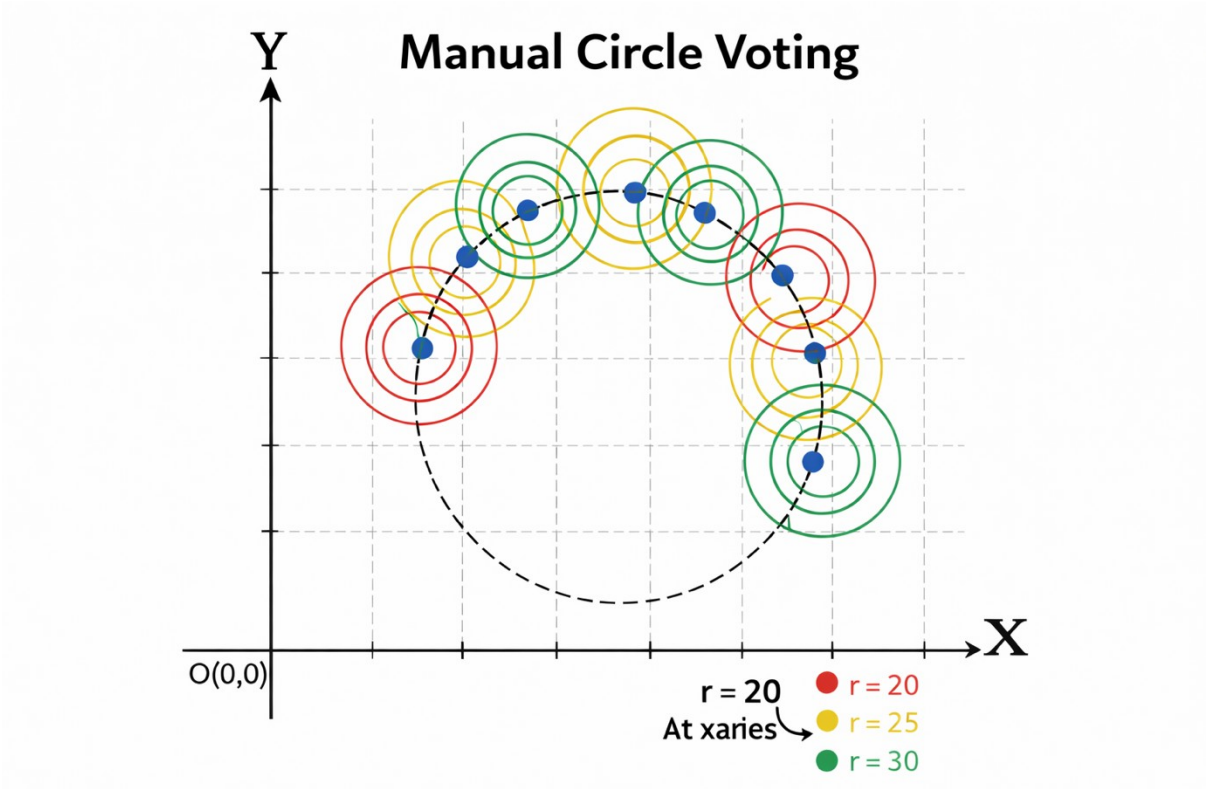
Unit 3 - Session 4

SLO 2: Evaluate computational challenges in circle detection.

Duration: 10 Minutes Activity

Question:



Q1.
What do the blue points on the diagram represent in the context of circle detection?

Answer: The blue points represent edge points detected in the image.

Q2.
Why are multiple circles drawn around each blue edge point?

Answer: Multiple circles represent different possible circle centers/radii voted for by each edge point.

Q3.
What do the different coloured circles indicate?

Answer: The different coloured circles indicate different radius values (r = 20, r = 25, and r = 30).

Q4.
How does increasing the number of radius values affect the voting process?

Answer: Increasing the number of radius values increases the number of possible votes and the computational work.

Q5.
Why does each edge point contribute a large number of votes in Circular Hough Transform?

Answer: Each edge point can vote for many possible circle centers for different radius values.

Q6.
What computational problem arises when many edge points are present in the image?

Answer: The number of votes and computations becomes very large, increasing processing time.

Q7.
How does the size of the accumulator space impact memory usage?

Answer: A larger accumulator space requires more memory to store votes for possible circle parameters.

Q8.
Why is full Circular Hough Transform often unsuitable for real-time applications?

Answer: It is computationally expensive because it requires many votes and a large accumulator space, making it slow for real-time use.